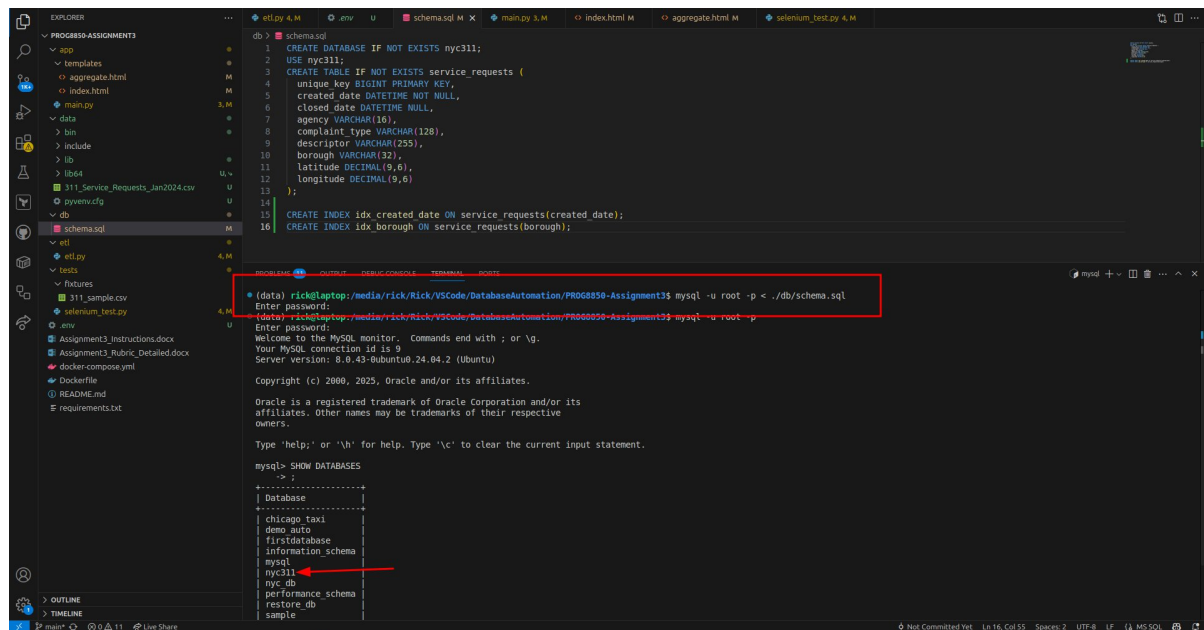


[illegible]

3. Database Schema & Indexes



The screenshot shows a VS Code editor with a file explorer on the left containing a project named 'PROG8850-Assignment3'. The main editor displays a file named 'schema.sql' with the following SQL code:

```
1 CREATE DATABASE IF NOT EXISTS nyc311;
2 USE nyc311;
3 CREATE TABLE IF NOT EXISTS service_requests (
4   unique_key BIGINT PRIMARY KEY,
5   created_date DATETIME NOT NULL,
6   closed_date DATETIME NULL,
7   agency VARCHAR(16),
8   complaint_type VARCHAR(128),
9   descriptor VARCHAR(255),
10  borough VARCHAR(32),
11  latitude DECIMAL(9,6),
12  longitude DECIMAL(9,6)
13 );
14
15 CREATE INDEX idx_created_date ON service_requests(created_date);
16 CREATE INDEX idx_borough ON service_requests(borough);
```

Below the editor, a terminal window shows the command to run the schema file:

```
rick@laptop:/media/rick/Rick/VSCode/DatabaseAutomation/PROG8850-Assignment3$ mysql -u root -p < ./db/schema.sql
```

The terminal output shows the MySQL prompt and the execution of the schema file:

```
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

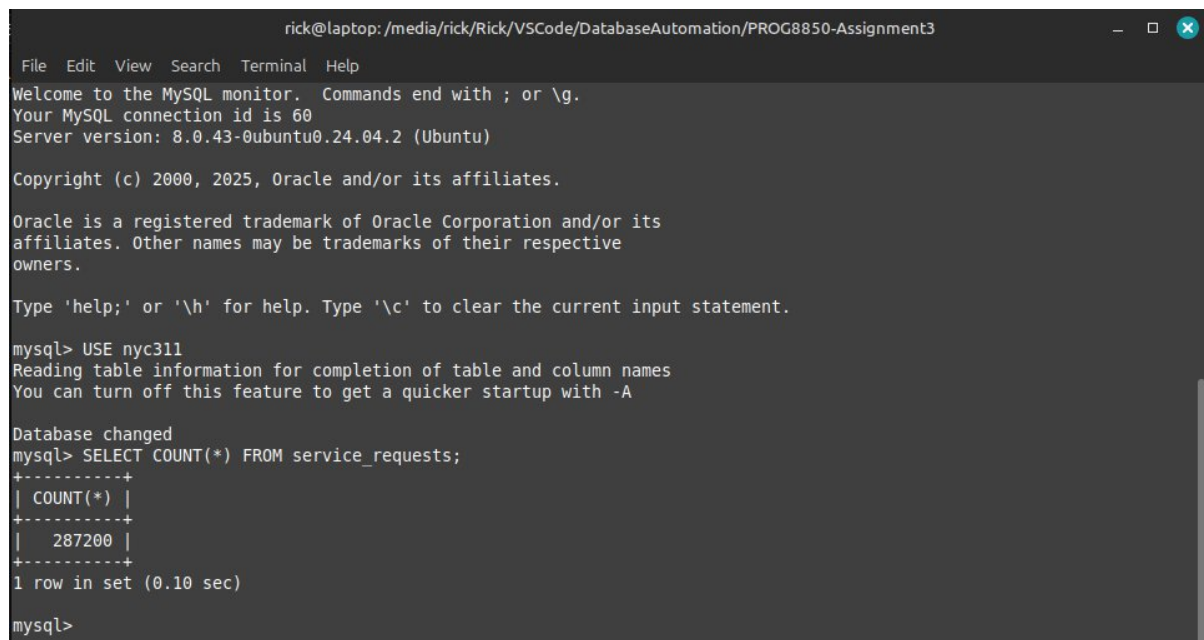
Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES
+-----+
| Database |
+-----+
| chicanco_taxi |
| demo_suito |
| firstdatabase |
| information_schema |
| mysql |
| nyc311 |
| nyc_db |
| performance_schema |
| restore_db |
| sample |
+-----+
```

I decided to implement index on a DATETIME column like created_date because it substantially speeds up range queries, as MySQL can use the index efficiently if the query uses direct comparisons or range operators without wrapping in functions like DATE(). I also implemented the borough index to help with selective lookups by borough.



The screenshot shows a terminal window with the following commands and output:

```
rick@laptop:/media/rick/Rick/VSCode/DatabaseAutomation/PROG8850-Assignment3

File Edit View Search Terminal Help

Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 60
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE nyc311
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT COUNT(*) FROM service_requests;
+-----+
| COUNT(*) |
+-----+
| 287200 |
+-----+
1 row in set (0.10 sec)

mysql>
```

4. Web Application

←→🔄🕒127.0.0.1:5000

☆🔴📧🔍⋮

NYC 311 Service Requests

Borough: Complaint Type: Start Date: End Date:

287200 results found.

Unique Key	Date	Agency	Complaint	Borough
60191146	2024-01-31 23:59:55	NYPD	Illegal Parking	BROOKLYN
60194494	2024-01-31 23:59:13	HPD	HEAT/HOT WATER	BRONX
60192234	2024-01-31 23:59:08	DOT	Street Condition	QUEENS
60200052	2024-01-31 23:59:05	TLC	Taxi Complaint	QUEENS
60199465	2024-01-31 23:59:03	NYPD	Noise - Residential	BROOKLYN
60193943	2024-01-31 23:59:00	DEP	Noise	MANHATTAN
60191450	2024-01-31 23:59:00	DEP	Noise	MANHATTAN
60193218	2024-01-31 23:58:53	HPD	HEAT/HOT WATER	BRONX
60194294	2024-01-31 23:58:51	NYPD	Noise - Commercial	MANHATTAN
60194291	2024-01-31 23:58:48	NYPD	Noise - Commercial	BROOKLYN
60196410	2024-01-31 23:58:35	HPD	HEAT/HOT WATER	QUEENS
60193370	2024-01-31 23:58:09	HPD	PLUMBING	MANHATTAN
60194299	2024-01-31 23:57:54	NYPD	Noise - Residential	BROOKLYN
60192627	2024-01-31 23:57:43	DSNY	Vendor Enforcement	QUEENS
60191015	2024-01-31 23:57:41	NYPD	Abandoned Vehicle	QUEENS
60198049	2024-01-31 23:57:38	NYPD	Blocked Driveway	BRONX
60194945	2024-01-31 23:55:57	NYPD	Noise - Residential	BRONX
60199361	2024-01-31 23:55:54	NYPD	Blocked Driveway	BROOKLYN
60196220	2024-01-31 23:55:51	NYPD	Noise - Street/Sidewalk	QUEENS
60199761	2024-01-31 23:55:44	HPD	HEAT/HOT WATER	BROOKLYN

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[View Complaints per Borough](#)

←→🔄🕒127.0.0.1:5000/?borough=bronx&complaint_type=&start_date=&end_date=

☆🔴📧🔍⋮

NYC 311 Service Requests

Borough: Complaint Type: Start Date: End Date:

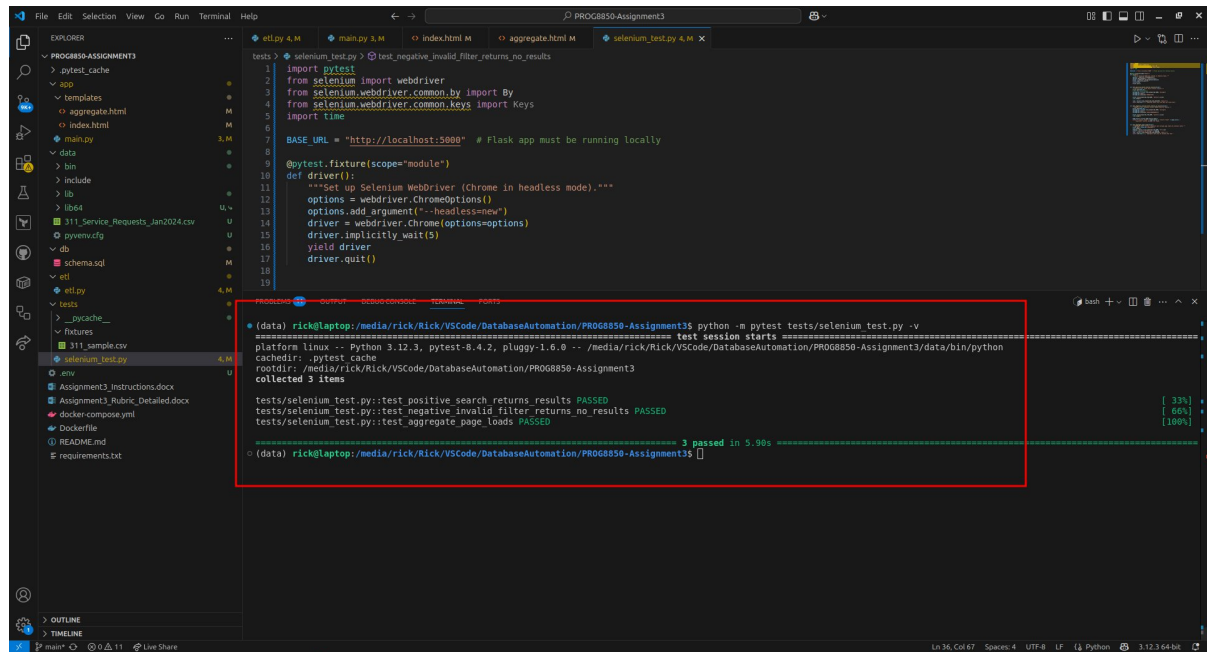
61455 results found.

Unique Key	Date	Agency	Complaint	Borough
60194494	2024-01-31 23:59:13	HPD	HEAT/HOT WATER	BRONX
60193218	2024-01-31 23:58:53	HPD	HEAT/HOT WATER	BRONX
60198049	2024-01-31 23:57:38	NYPD	Blocked Driveway	BRONX
60194945	2024-01-31 23:55:57	NYPD	Noise - Residential	BRONX
60195576	2024-01-31 23:55:37	NYPD	Noise - Residential	BRONX
60195593	2024-01-31 23:51:17	NYPD	Noise - Residential	BRONX
60198791	2024-01-31 23:51:00	NYPD	Noise - Commercial	BRONX
60210084	2024-01-31 23:50:47	DOB	General Construction/Plumbing	BRONX
60197997	2024-01-31 23:50:25	NYPD	Noise - Residential	BRONX
60199760	2024-01-31 23:49:01	HPD	HEAT/HOT WATER	BRONX
60192753	2024-01-31 23:45:51	NYPD	Illegal Parking	BRONX
60193206	2024-01-31 23:42:03	HPD	HEAT/HOT WATER	BRONX
60196197	2024-01-31 23:39:46	NYPD	Illegal Parking	BRONX
60199788	2024-01-31 23:36:25	HPD	PAINT/PLASTER	BRONX
60196990	2024-01-31 23:36:25	HPD	FLOORING/STAIRS	BRONX
60195875	2024-01-31 23:36:25	HPD	PAINT/PLASTER	BRONX
60192220	2024-01-31 23:36:25	HPD	DOOR/WINDOW	BRONX
60190826	2024-01-31 23:36:25	HPD	PAINT/PLASTER	BRONX
60195398	2024-01-31 23:32:00	DEP	Noise	BRONX
60199467	2024-01-31 23:30:38	NYPD	Noise - Residential	BRONX

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[View Complaints per Borough](#)

5. Automated testing



The screenshot shows a VS Code editor with a project named 'PROG8850-Assignment3'. The file explorer on the left shows a directory structure with files like 'selenium_test.py', 'main.py', 'index.html', and 'aggregate.html'. The main editor displays the content of 'selenium_test.py', which is a Python script using pytest and Selenium WebDriver to test a web application. The script defines a fixture for the WebDriver and contains three test functions: 'test_positive_search_returns_results', 'test_negative_invalid_filter_returns_no_results', and 'test_aggregate_page_loads'. The terminal at the bottom shows the command 'python -m pytest tests/selenium_test.py -v' being executed, resulting in three passed tests.

```
tests > selenium_test.py > test_negative_invalid_filter_returns_no_results
1 import pytest
2 from selenium import webdriver
3 from selenium.webdriver.common.by import By
4 from selenium.webdriver.common.keys import Keys
5 import time
6
7 BASE_URL = "http://localhost:5000" # Flask app must be running locally
8
9 @pytest.fixture(scope="module")
10 def driver():
11     """Set up Selenium WebDriver (Chrome in headless mode)."""
12     options = webdriver.ChromeOptions()
13     options.add_argument("--headless=new")
14     driver = webdriver.Chrome(options=options)
15     driver.implicitly_wait(5)
16     yield driver
17     driver.quit()
18
19
20 tests/selenium_test.py:10 test_positive_search_returns_results PASSED [ 33%]
tests/selenium_test.py:15 test_negative_invalid_filter_returns_no_results PASSED [ 66%]
tests/selenium_test.py:20 test_aggregate_page_loads PASSED [100%]
===== 3 passed in 5.90s =====
(data) rick@laptop:/media/ric/Rick/VSCode/DatabaseAutomation/PROG8850-Assignment3$
```

6. GitHub Repository Link

Provide the URL to your GitHub repository:

<https://github.com/Richard-Conestoga/dbautomation-assignment3.git>